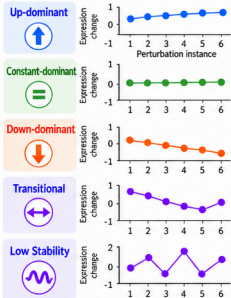


Figure 1. Feature Signal Stratification

From perturbation effects to condition-level signal architecture

a Signal behaviors under perturbation

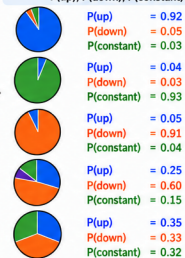
A feature is perturbed repeatedly; its response pattern is recorded across perturbations.



b Probabilistic representation in FSF state space

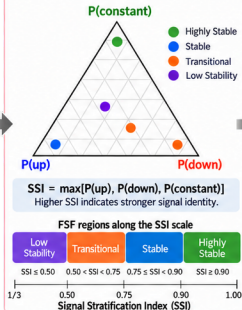
For each feature, the proportions of up, down and constant responses are estimated.

State probabilities:
 $P(\text{up})$, $P(\text{down})$, $P(\text{constant})$



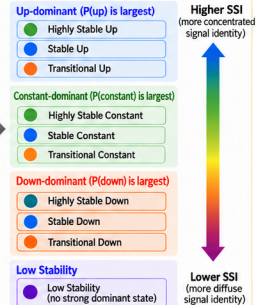
c Signal Stratification Index (SSI) quantifies signal identity concentration

Using state probabilities, FSF computes the Signal Stratification Index (SSI).



d Signal stratification into FSF classes

Features are assigned to FSF classes based on dominant state and SSI region.



e Condition-level signal architecture

Aggregating FSF classes across features reveals the signal architecture of each condition.

Highly Stable-dominated architecture (e.g., HT, LT, OSM)



Two-region architecture (e.g., DES)



Multi-layer architecture (e.g., UV, GAM)



FSF class colors (legend)



FSF characterizes feature responses under perturbation, represents them in a three-state probability space, quantifies signal identity with SSI, stratifies features into interpretable FSF classes, and reveals condition-level signal architecture.